

Computational Learning Theory

Computational Learning Theory (CoLT)

- **Definition:** Computational Learning Theory is the mathematical study of machine learning.
- It provides the **theoretical foundation of machine learning** by studying the conditions under which learning is possible, the amount of data required, the computational resources needed, and the ability of learned models to generalize.
- It answers questions such as:
 - Can a machine learn a concept?
 - How many training samples are needed?
 - How accurate will the learned model be?
 - How much computation is required?
- It combines ideas from:
 - Computer Science
 - Probability
 - Statistics
 - Mathematics

- **Example**

- Suppose we want a computer to identify emails as **Spam** or **Not Spam**.

- Training data:

Email	Label
• Win money now	Spam
• Meeting at 5 PM	Not Spam
• Free lottery	Spam

- CoLT studies:

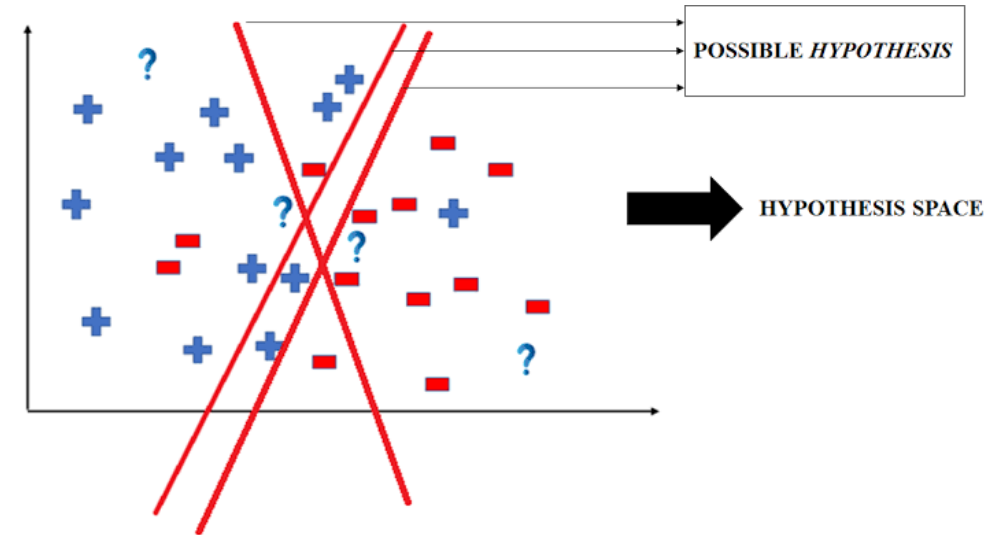
- How many emails are enough?
 - Will the learned classifier work on unseen emails?
 - How likely is it to make mistakes?

Key Concepts OF CoLT

- **1. Learning Model**

- A learning model consists of:

- **Training examples:** Input-output pairs used for learning.
- **Hypothesis space (H):** Set of all possible models.
- **Learning algorithm:** Chooses the best hypothesis based on training data.
- **Target concept:** The true function that generated the data.



2. PAC Learning (Probably Approximately Correct)

- Introduced by Leslie Valiant in 1984, PAC learning answers the question: *When can an algorithm reliably generalize from training data?*
- In PAC learning, an algorithm receives a sample of labeled data drawn from an unknown distribution. The goal isn't to find a "perfect" model, but rather a model that is:
 - **Approximately Correct:** The model's error rate is less than a small threshold ϵ (epsilon).
 - **Probably correct:** The probability that the model fails to achieve this low error rate is less than a small confidence parameter δ (delta).

- Mathematically,

$$P(\text{Error}(h) \leq \epsilon) \geq 1 - \delta$$

where:

- h = learned hypothesis
- ϵ = acceptable error
- δ = probability of failure

Example

Suppose

$$\epsilon = 0.05$$

$$\delta = 0.01$$

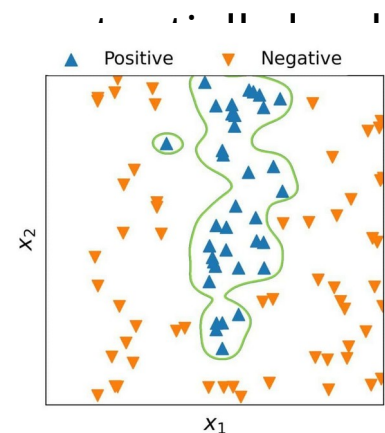
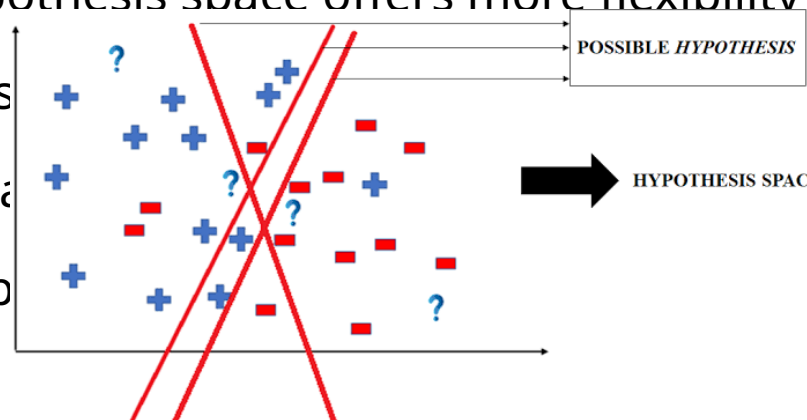
This means

The learner finds a hypothesis with:

- error $\leq 5\%$
- with probability 99%

3. Hypothesis Space

- The hypothesis space (H) is the set of all possible hypotheses (or models) that a learning algorithm can choose from.
- A larger hypothesis space can represent more complex functions but usually requires more training data.
- The learning algorithm evaluates these hypotheses using the training data and selects the one with the lowest error.
- In PAC learning, the size and structure of the hypothesis space play a crucial role in determining the sample complexity and the generalization ability of the algorithm.
- A larger and more complex hypothesis space offers more flexibility and can represent more accurate models. However, it also runs the risk of overfitting to the training data. Therefore, the learning algorithm must balance the flexibility of the hypothesis space with the need for generalization.



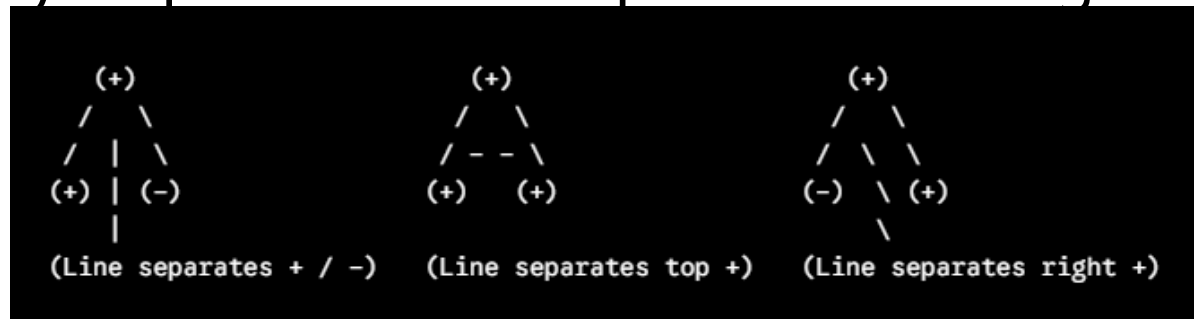
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- **4. Generalization**

- Generalization is the ability of a model to perform well on **new, unseen data** rather than memorizing the training data.
- A good learning algorithm minimizes both:
 - Training error
 - Test (generalization) error

5. VC Dimension

- The **Vapnik-Chervonenkis (VC) Dimension** is a measure of the capacity or complexity of the hypothesis space.
- It quantifies the maximum number of points that can be shattered (i.e., correctly classified in all possible ways) by the hypotheses in the space.
- A higher VC dimension indicates a more complex hypothesis space, which may require more samples to ensure good generalization.



6. Sample Complexity

- **Definition**

- Sample complexity is the minimum number of training examples required to learn a concept with high confidence and low error.

- Simply put:

- "How much data do we need before learning becomes reliable?"

- Intuition

- Suppose you want to recognize cats.
- With only **3 pictures**, your model may perform poorly.
- With **10,000 pictures**, the model learns much better.
- The number of examples needed is the sample complexity.

- For PAC learning,

$$m = O\left(\frac{VC(H) \log(1/\epsilon) + \log(1/\delta)}{\epsilon}\right)$$

where:

- $VC(H)$ = VC dimension
- ϵ = allowable error
- δ = confidence parameter

Example

Suppose

- $|H| = 1000$
- error = 5%
- confidence = 95%

The formula estimates how many examples are sufficient.

ϵ -Exhausted Version Space

Version Space

Version space is

The set of all hypotheses consistent with the observed training examples.

Suppose

Hypotheses:

- h_1
- h_2
- h_3
- h_4

Training data eliminates h_2 and h_4 .

Remaining hypotheses:

- h_1
- h_3

This remaining set is the **version space**.

ϵ -Exhausted Version Space

A version space is ϵ -exhausted if every hypothesis remaining in the version space has true error at most ϵ .

Meaning:

No "bad" hypothesis with error greater than ϵ remains.

Only good hypotheses survive.

Intuition

Suppose

Actual classifier:

✓ Cats

Remaining hypotheses:

Hypothesis	Error
h_1	2%
h_2	3%
h_3	4%

If

$$\epsilon = 5\%$$

All errors are below ϵ .

Therefore,

Version Space is **ϵ -exhausted**.

Agnostic Learner

Agnostic learning is a learning framework in which **the algorithm does not assume that the training data can be perfectly explained by any hypothesis in the hypothesis space**. In other words, the data may contain **noise, errors, or inconsistencies**, so there may not be a "correct" hypothesis.

Unlike PAC learning, which assumes that a perfect target function exists within the hypothesis space, agnostic learning makes **no such assumption**.

Definition

In agnostic learning, the goal is to find the hypothesis that has the **smallest possible prediction error** among all hypotheses in the hypothesis space.

If H is the hypothesis space, the learner seeks

$$h^* = \arg \min_{h \in H} \text{Error}(h)$$

That is, it chooses the hypothesis with the **minimum error**, even if that error is not zero.

Difference

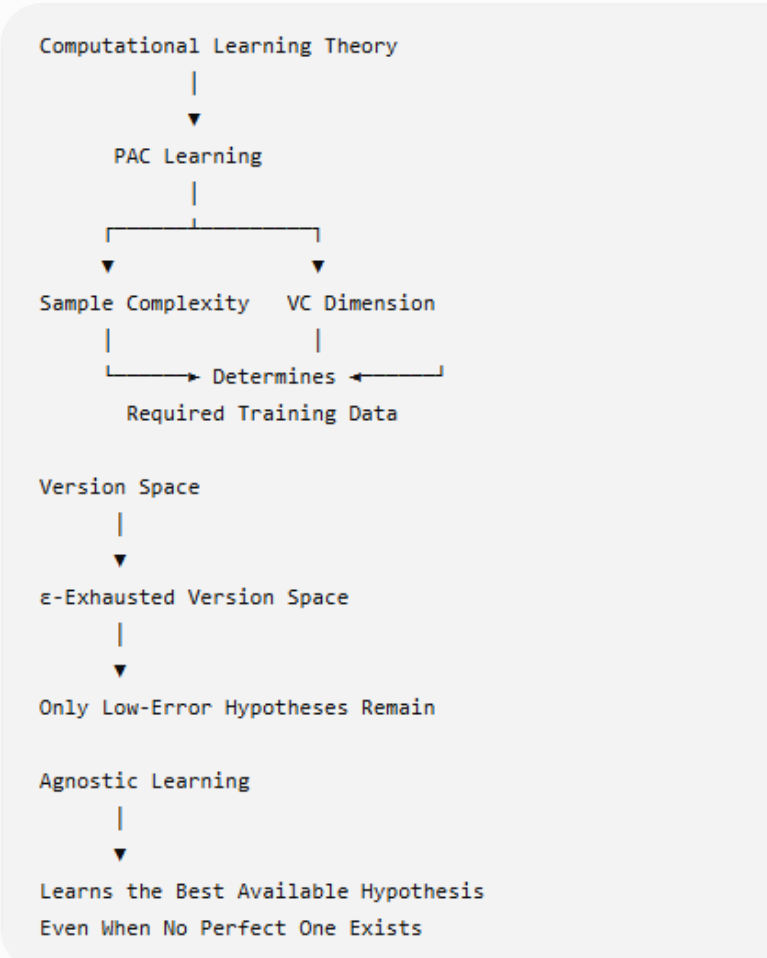
PAC Learning:

Perfect hypothesis exists

Agnostic Learning:

Perfect hypothesis may not exist
Choose the best available hypothesis

Relationship Among These Concepts



Concept	Meaning	Key Idea
Computational Learning Theory	Mathematical framework for machine learning	Studies what can be learned and how efficiently
Sample Complexity	Number of training examples needed	More complex problems require more data
Version Space	Set of hypotheses consistent with training data	Contains all hypotheses not yet ruled out
ε-Exhausted Version Space	Version space where every remaining hypothesis has error $\leq \epsilon$	Only sufficiently accurate hypotheses remain
PAC Learning	Learn with bounded error and high confidence	Error $\leq \epsilon$ with probability $\geq 1 - \delta$
Agnostic Learner	Learns without assuming a perfect hypothesis exists	Finds the hypothesis with the lowest achievable error
VC Dimension	Measure of hypothesis class complexity	Higher VC dimension means greater expressive power and typically higher sample